	Parul University **Faculty of Engineering and Technology** **Parul Institute of Engineering and Technology** **Computer Science & Engineering Department**

Subject Name	**Operating System**	**A.Y**	**2025-26**
Subject Code	**303105251**	**Semester**	**4^{th} sem**

Chapter-4

Sr No	Question	COs	B.T
1	What is Circular Wait?	4	1
2	Define Mutual Exclusion in context of deadlock.	4	1
3	Differentiate between deadlock prevention and avoidance.	4	2
4	Describe Deadlock in detail. Discuss prevention, avoidance, detection, and recovery.	4	2
5	Show how the Banker’s algorithm determines a safe sequence.	4	3
6	Explain deadlock detection algorithms and compare their complexity.	4	3
7	With a neat diagram, explain safe and unsafe states in deadlock avoidance.	4	3
8	Explain the Banker’s Algorithm for safety and resource allocation with an example.	4	3
9	What is the basic idea of Banker’s Algorithm?	4	2
10	What is No Preemption?	4	1
11	What are the four necessary conditions for deadlock?	4	1
12	Give two methods of deadlock recovery.	4	2
13	Explain deadlock handling methods with advantages and disadvantages.	4	3
14	Explain resource allocation graph (RAG) and its role in detecting deadlocks.	4	3
15	Explain the different recovery strategies in deadlock.	4	3

Unit - 5

Sr No	Question	COs	B.T
1	Define Memory Management.	5	1
2	What is internal fragmentation?	5	1
3	What is a page table?	5	1
4	What is paging?	5	1
5	What is a page fault?	5	1
6	What is memory fragmentation?	5	1
7	What is a dirty bit?	5	1
8	Explain FIFO page replacement with example.	5	2
9	Explain the Optimal page replacement algorithm.	5	3
10	Explain LRU page replacement with example.	5	3
11	Explain fixed and variable partition allocation.	5	2
12	Describe demand paging, page faults, dirty pages, and handling mechanism.	5	3

13	Describe fixed and variable partition allocation. Compare both.	**5**	**4**
14	Explain hardware support required for paging.	**5**	**2**
15	What is demand paging?	**5**	**2**
16	Explain internal and external fragmentation with examples.	**5**	**2**
17	Explain compaction. When is it needed?	**5**	**2**
18	Explain second-chance (clock) page replacement algorithm	**5**	**3**
19	Explain FIFO, LRU, Optimal, Second Chance, and NRU algorithms.	**5**	**3**
20	Compare paging and segmentation.	**5**	**3**